Quiz 1 - Part 1: Discrete Mathematics (CSE121)

Date: 01-09-2025 **Maximum marks**: 40

Instructions. Please read the following points carefully.

- Mention your names, roll numbers, and the part number of the quiz in your answer sheets.
- Do not use your mobile phones, tablets, laptops, books, notebooks, or any other material during the exam.
- Do not talk to others or look at someone's answer sheet. If you have any questions, please ask the instructor or TAs.
- This exam is designed for 50 minutes, with four questions, each worth 10 points.

1 Problem Set

In this document, $\equiv$ denotes equivalence of propositions.

Question 1.1. 1. *(5 marks) Let p, q, r be propositions. Write the truth table of the following propositions.*

(a) $p \implies q \wedge \neg q \iff r$.

(b) $(p \oplus \neg q) \vee (r \implies p)$.

Answer hint: Apply the precedence order of logical operators.

2. *(5 marks). Prove that if two propositions are equivalent then they have the same DNFs.* *Answer hint: There are two ways of doing this - one by truth tables and the other by the algebra of propositional logic. In the former method, we know that two equivalent propositions have the same truth table. Following the procedure to obtain a DNF equivalent to a proposition from its truth table, we get that two equivalent propositions have the same DNFs.*

Question 1.2. 1. *(6 marks). State whether the following statements are true or false. Briefly justify your answers.*

(a) *"How are you?" is a proposition.*
Answer: Questions are not declarative statements, and hence not propositions.

(b) $p \implies q \equiv \neg p \vee \neg q$, *where p, q are propositional variables.*
Answer hint: Verify using truth tables.

(c) $(p \vee q) \wedge (\neg p \wedge \neg q)$ *is a contingency, where* p, q *are propositional variables.*
Answer hint: False. Try all possible truth values.

(d) *Let* x *be a variable and* $P(x)$ *be the predicate* $x^2 \geq x + 3$. *Suppose the domain of* $P(x)$ *is all natural numbers. Then,* $\forall x P(x)$ *is true.*
Answer hint: Use the table taught in the class that says when $\forall x P(x)$ *is true.*

2. *(4 marks). Let* x, y *be variables and* $P(x, y)$ *be the predicate* $x^2 + 3 \leq x^3, y^2 - 1 \geq y$. *Suppose the domains of* x, y *in* $P(x, y)$ *are all the natural numbers. What will be the truth value of* $\exists x \, \forall y P(x, y)$*?*
Answer hint: $P(x, y)$ *is interpreted as* $x^2 + 3 \leq x^3 \wedge y^2 - 1 \geq y$. *In* $\exists x \forall y P(x, y)$*, first apply the inner quantifier, i.e.,* $\forall y$ *and then apply* $\exists x$. *In this case, take* $x = 3$ *and* $y = 1$. *Then,* $P(x = 3, y = 1)$ *is false.*

Question 1.3. *Let* p, q, r *be propositions* $u := \neg(p \oplus q) \iff \neg r$.

1. *(8 marks). Construct a CNF* v *and a DNF* w *such that* $v \equiv u$ *and* $u \equiv w$.
Answer hint: Can be done either using the truth-table approach or using the rules of the algebra of propositional logic. Both are done in class.

2. *(2 marks). Let* p *and* q *denote the statements "Twin-prime conjecture is true" and "2024 is a leap year". We know that* p *is a conjecture. Can we still infer the truth value of* $p \implies q$*?*
Answer hint: Use the truth table of $\implies$.

Question 1.4. 1. *(4 marks). Let* x, y *be variables and* $P(x, y), Q(x, y)$ *be propositions. Simplify the following using rules of predicate and propositional logic*

$$\neg \exists x \exists y (P(x, y) \vee Q(x, y)).$$

Answer hint: Apply the DeMorgan's laws for quantifiers and logical operations.

2. *(4 marks). Do we need quantifiers if we only have predicates with finitely many elements in their domains? Justify your answer.*
Answer: No. Done in the class.

3. *(2 marks). Mention your favourite topic in mathematical logic.*